

Multiple Choice (10 marks)

1. 5.8-5. If the distribution of Y is $b(n, 0.25)$, give a lower bound for $P(|Y/n - 0.25| < 0.05)$ when (a) $n = 100$.
 - (a) 0.75
 - (b) 0.85
 - (c) 0.45
 - (d) 0.35
 - (e) 0.55
2. In how many ways can 3 students be selected from a class of 20 to form a study group?
 - (a) 560
 - (b) 1140
 - (c) 1000
 - (d) 680
 - (e) 760
3. Use rounding to estimate the sum of $27.94 + 15.35$ to the nearest tenth.
 - (a) 44
 - (b) 42.9
 - (c) 44.1
 - (d) 42.8
 - (e) 43.29
4. Which of the following is an equation of the line tangent to the graph of $y = x + e^x$ at $x = 0$?
 - (a) $y = x + 3$
 - (b) $y = x + 2$
 - (c) $y = 2x$
 - (d) $y = x - 1$
 - (e) $y = x - 2$
5. What is the smallest positive integer n such that $\frac{1}{n}$ is a terminating decimal and n contains the digit 9?
 - (a) 512

- (b) 1024
- (c) 128
- (d) 72
- (e) 32

True / False (10 marks)

Answer True or False

6. True or False: A regular octagon has 12 diagonals, which is derived from a common misunderstanding of diagonal counting.
7. True or False: Tayshawn can sort 56 marbles into 9 groups with 6 marbles in each group, as 9 multiplied by 6 equals 54.
8. True or False: If Hardy picked the complex number $5+3i$, then Ramanujan could have picked $4-i$ to achieve the product $32-8i$.
9. True or False: The value of c in $\mathbb{Z}_3$ that makes $\mathbb{Z}_3[x]/(x^3 + cx^2 + 1)$ a field is 1.
10. True or False: The mixed number representation of 1 and 1 over 11 is equivalent to 2 over 11.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Let z be a complex number such that

$$z^2 + |z|^2 = 3 - 5i.$$

Find $|z|^2$.

12. Calculate and discuss the elevation difference between Salt Flats, at -55 feet, and Talon Bluff, at 620 feet. Explain the mathematical reasoning behind your solution.

End of Paper